Lesson Planning

Complete Lesson Plan Book

Class: Class 12 – FSc Pre-Medical

Subject: Biology

Medium: English

Academic Year 2026-27

DM

This book is prepared as a teaching guide for teachers.

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Unit 1: Homeostasis

Learning Objectives

• Define and explain water balance in the context of Homeostasis.
• Differentiate between kidney function and dialysis with suitable examples.
• Solve conceptual problems related to diabetes accurately.
• Apply the ideas of Homeostasis to everyday situations involving feedback mechanism.

Key Concepts

• Meaning, definition and scope of water balance.
• Important properties, types and examples of kidney function.
• Relationship between dialysis and diabetes.
• Practical applications of Homeostasis, especially feedback mechanism.
• Key terms and a summary of the main points of Homeostasis.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Homeostasis with a daily-life example.
2. Explain water balance on the board with definitions and clear diagrams.
3. Discuss kidney function and dialysis using simple, relatable examples.
4. Involve the students through short questions on diabetes.
5. Solve an exercise question on feedback mechanism together with the class.
6. Summarize the lesson and highlight the key points of Homeostasis.

Classroom Activities

• Group discussion: students compare kidney function with dialysis.
• Board activity: solve a question on diabetes in pairs.
• Quick quiz on the key terms of Homeostasis.

Assessment Questions

1. Define water balance.
2. Differentiate between kidney function and dialysis.
3. Explain with an example: diabetes.
4. Write a short note on feedback mechanism.

Homework

• Read Homeostasis from the book and solve the exercise questions.
• Prepare short notes on water balance and kidney function.

Time Allocation

4–6 periods of 40 minutes each

Unit 2: Coordination and Control

Learning Objectives

• Define and explain brain structure in the context of Coordination and Control.
• Differentiate between spinal cord and endocrine glands with suitable examples.
• Solve conceptual problems related to diabetes mellitus accurately.
• Apply the ideas of Coordination and Control to everyday situations involving nervous disorders.

Key Concepts

• Meaning, definition and scope of brain structure.
• Important properties, types and examples of spinal cord.
• Relationship between endocrine glands and diabetes mellitus.
• Practical applications of Coordination and Control, especially nervous disorders.
• Key terms and a summary of the main points of Coordination and Control.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Coordination and Control with a daily-life example.
2. Explain brain structure on the board with definitions and clear diagrams.
3. Discuss spinal cord and endocrine glands using simple, relatable examples.
4. Involve the students through short questions on diabetes mellitus.
5. Solve an exercise question on nervous disorders together with the class.
6. Summarize the lesson and highlight the key points of Coordination and Control.

Classroom Activities

• Group discussion: students compare spinal cord with endocrine glands.
• Board activity: solve a question on diabetes mellitus in pairs.
• Quick quiz on the key terms of Coordination and Control.

Assessment Questions

1. Define brain structure.
2. Differentiate between spinal cord and endocrine glands.
3. Explain with an example: diabetes mellitus.
4. Write a short note on nervous disorders.

Homework

• Read Coordination and Control from the book and solve the exercise questions.
• Prepare short notes on brain structure and spinal cord.

Time Allocation

4–6 periods of 40 minutes each

Unit 3: Reproduction

Learning Objectives

• Define and explain gametogenesis in the context of Reproduction.
• Differentiate between human embryo development and plant life cycle with suitable examples.
• Solve conceptual problems related to parthenogenesis accurately.
• Apply the ideas of Reproduction to everyday situations involving contraception.

Key Concepts

• Meaning, definition and scope of gametogenesis.
• Important properties, types and examples of human embryo development.
• Relationship between plant life cycle and parthenogenesis.
• Practical applications of Reproduction, especially contraception.
• Key terms and a summary of the main points of Reproduction.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Reproduction with a daily-life example.
2. Explain gametogenesis on the board with definitions and clear diagrams.
3. Discuss human embryo development and plant life cycle using simple, relatable examples.
4. Involve the students through short questions on parthenogenesis.
5. Solve an exercise question on contraception together with the class.
6. Summarize the lesson and highlight the key points of Reproduction.

Classroom Activities

• Group discussion: students compare human embryo development with plant life cycle.
• Board activity: solve a question on parthenogenesis in pairs.
• Quick quiz on the key terms of Reproduction.

Assessment Questions

1. Define gametogenesis.
2. Differentiate between human embryo development and plant life cycle.
3. Explain with an example: parthenogenesis.
4. Write a short note on contraception.

Homework

• Read Reproduction from the book and solve the exercise questions.
• Prepare short notes on gametogenesis and human embryo development.

Time Allocation

4–6 periods of 40 minutes each

Unit 4: Support and Movement

Learning Objectives

• Define and explain human skeleton in the context of Support and Movement.
• Differentiate between joints and muscles with suitable examples.
• Solve conceptual problems related to muscle contraction accurately.
• Apply the ideas of Support and Movement to everyday situations involving plant movements.

Key Concepts

• Meaning, definition and scope of human skeleton.
• Important properties, types and examples of joints.
• Relationship between muscles and muscle contraction.
• Practical applications of Support and Movement, especially plant movements.
• Key terms and a summary of the main points of Support and Movement.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Support and Movement with a daily-life example.
2. Explain human skeleton on the board with definitions and clear diagrams.
3. Discuss joints and muscles using simple, relatable examples.
4. Involve the students through short questions on muscle contraction.
5. Solve an exercise question on plant movements together with the class.
6. Summarize the lesson and highlight the key points of Support and Movement.

Classroom Activities

• Group discussion: students compare joints with muscles.
• Board activity: solve a question on muscle contraction in pairs.
• Quick quiz on the key terms of Support and Movement.

Assessment Questions

1. Define human skeleton.
2. Differentiate between joints and muscles.
3. Explain with an example: muscle contraction.
4. Write a short note on plant movements.

Homework

• Read Support and Movement from the book and solve the exercise questions.
• Prepare short notes on human skeleton and joints.

Time Allocation

4–6 periods of 40 minutes each

Unit 5: Growth and Development

Learning Objectives

• Define and explain growth phases in the context of Growth and Development.
• Differentiate between plant growth regulators and human growth stages with suitable examples.
• Solve conceptual problems related to aging accurately.
• Apply the ideas of Growth and Development to everyday situations involving abnormal growth.

Key Concepts

• Meaning, definition and scope of growth phases.
• Important properties, types and examples of plant growth regulators.
• Relationship between human growth stages and aging.
• Practical applications of Growth and Development, especially abnormal growth.
• Key terms and a summary of the main points of Growth and Development.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Growth and Development with a daily-life example.
2. Explain growth phases on the board with definitions and clear diagrams.
3. Discuss plant growth regulators and human growth stages using simple, relatable examples.
4. Involve the students through short questions on aging.
5. Solve an exercise question on abnormal growth together with the class.
6. Summarize the lesson and highlight the key points of Growth and Development.

Classroom Activities

• Group discussion: students compare plant growth regulators with human growth stages.
• Board activity: solve a question on aging in pairs.
• Quick quiz on the key terms of Growth and Development.

Assessment Questions

1. Define growth phases.
2. Differentiate between plant growth regulators and human growth stages.
3. Explain with an example: aging.
4. Write a short note on abnormal growth.

Homework

• Read Growth and Development from the book and solve the exercise questions.
• Prepare short notes on growth phases and plant growth regulators.

Time Allocation

4–6 periods of 40 minutes each

Unit 6: Chromosomes and DNA

Learning Objectives

• Define and explain chromosome structure in the context of Chromosomes and DNA.
• Differentiate between DNA structure and DNA replication with suitable examples.
• Solve conceptual problems related to RNA types accurately.
• Apply the ideas of Chromosomes and DNA to everyday situations involving protein synthesis.

Key Concepts

• Meaning, definition and scope of chromosome structure.
• Important properties, types and examples of DNA structure.
• Relationship between DNA replication and RNA types.
• Practical applications of Chromosomes and DNA, especially protein synthesis.
• Key terms and a summary of the main points of Chromosomes and DNA.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Chromosomes and DNA with a daily-life example.
2. Explain chromosome structure on the board with definitions and clear diagrams.
3. Discuss DNA structure and DNA replication using simple, relatable examples.
4. Involve the students through short questions on RNA types.
5. Solve an exercise question on protein synthesis together with the class.
6. Summarize the lesson and highlight the key points of Chromosomes and DNA.

Classroom Activities

• Group discussion: students compare DNA structure with DNA replication.
• Board activity: solve a question on RNA types in pairs.
• Quick quiz on the key terms of Chromosomes and DNA.

Assessment Questions

1. Define chromosome structure.
2. Differentiate between DNA structure and DNA replication.
3. Explain with an example: RNA types.
4. Write a short note on protein synthesis.

Homework

• Read Chromosomes and DNA from the book and solve the exercise questions.
• Prepare short notes on chromosome structure and DNA structure.

Time Allocation

4–6 periods of 40 minutes each

Unit 7: Variations and Genetics

Learning Objectives

• Define and explain Mendel's laws in the context of Variations and Genetics.
• Differentiate between monohybrid cross and dihybrid cross with suitable examples.
• Solve conceptual problems related to sex determination accurately.
• Apply the ideas of Variations and Genetics to everyday situations involving genetic disorders.

Key Concepts

• Meaning, definition and scope of Mendel's laws.
• Important properties, types and examples of monohybrid cross.
• Relationship between dihybrid cross and sex determination.
• Practical applications of Variations and Genetics, especially genetic disorders.
• Key terms and a summary of the main points of Variations and Genetics.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Variations and Genetics with a daily-life example.
2. Explain Mendel's laws on the board with definitions and clear diagrams.
3. Discuss monohybrid cross and dihybrid cross using simple, relatable examples.
4. Involve the students through short questions on sex determination.
5. Solve an exercise question on genetic disorders together with the class.
6. Summarize the lesson and highlight the key points of Variations and Genetics.

Classroom Activities

• Group discussion: students compare monohybrid cross with dihybrid cross.
• Board activity: solve a question on sex determination in pairs.
• Quick quiz on the key terms of Variations and Genetics.

Assessment Questions

1. Define Mendel's laws.
2. Differentiate between monohybrid cross and dihybrid cross.
3. Explain with an example: sex determination.
4. Write a short note on genetic disorders.

Homework

• Read Variations and Genetics from the book and solve the exercise questions.
• Prepare short notes on Mendel's laws and monohybrid cross.

Time Allocation

4–6 periods of 40 minutes each

Unit 8: Biotechnology

Learning Objectives

• Define and explain recombinant DNA in the context of Biotechnology.

• Differentiate between gene cloning and PCR with suitable examples.

• Solve conceptual problems related to transgenic organisms accurately.

• Apply the ideas of Biotechnology to everyday situations involving medical applications.

Key Concepts

• Meaning, definition and scope of recombinant DNA.

• Important properties, types and examples of gene cloning.

• Relationship between PCR and transgenic organisms.

• Practical applications of Biotechnology, especially medical applications.

• Key terms and a summary of the main points of Biotechnology.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Biotechnology with a daily-life example.
2. Explain recombinant DNA on the board with definitions and clear diagrams.
3. Discuss gene cloning and PCR using simple, relatable examples.
4. Involve the students through short questions on transgenic organisms.
5. Solve an exercise question on medical applications together with the class.
6. Summarize the lesson and highlight the key points of Biotechnology.

Classroom Activities

• Group discussion: students compare gene cloning with PCR.

• Board activity: solve a question on transgenic organisms in pairs.

• Quick quiz on the key terms of Biotechnology.

Assessment Questions

1. Define recombinant DNA.
2. Differentiate between gene cloning and PCR.
3. Explain with an example: transgenic organisms.
4. Write a short note on medical applications.

Homework

• Read Biotechnology from the book and solve the exercise questions.

• Prepare short notes on recombinant DNA and gene cloning.

Time Allocation

4–6 periods of 40 minutes each

Unit 9: Evolution

Learning Objectives

• Define and explain theories of evolution in the context of Evolution.
• Differentiate between natural selection and evidences of evolution with suitable examples.
• Solve conceptual problems related to human evolution accurately.
• Apply the ideas of Evolution to everyday situations involving speciation.

Key Concepts

• Meaning, definition and scope of theories of evolution.
• Important properties, types and examples of natural selection.
• Relationship between evidences of evolution and human evolution.
• Practical applications of Evolution, especially speciation.
• Key terms and a summary of the main points of Evolution.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Evolution with a daily-life example.
2. Explain theories of evolution on the board with definitions and clear diagrams.
3. Discuss natural selection and evidences of evolution using simple, relatable examples.
4. Involve the students through short questions on human evolution.
5. Solve an exercise question on speciation together with the class.
6. Summarize the lesson and highlight the key points of Evolution.

Classroom Activities

• Group discussion: students compare natural selection with evidences of evolution.
• Board activity: solve a question on human evolution in pairs.
• Quick quiz on the key terms of Evolution.

Assessment Questions

1. Define theories of evolution.
2. Differentiate between natural selection and evidences of evolution.
3. Explain with an example: human evolution.
4. Write a short note on speciation.

Homework

• Read Evolution from the book and solve the exercise questions.
• Prepare short notes on theories of evolution and natural selection.

Time Allocation

4–6 periods of 40 minutes each

Unit 10: Man and His Environment

Learning Objectives

• Define and explain ecosystem in the context of Man and His Environment.

• Differentiate between food chains and pollution with suitable examples.

• Solve conceptual problems related to conservation accurately.

• Apply the ideas of Man and His Environment to everyday situations involving global warming.

Key Concepts

• Meaning, definition and scope of ecosystem.

• Important properties, types and examples of food chains.

• Relationship between pollution and conservation.

• Practical applications of Man and His Environment, especially global warming.

• Key terms and a summary of the main points of Man and His Environment.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Man and His Environment with a daily-life example.
2. Explain ecosystem on the board with definitions and clear diagrams.
3. Discuss food chains and pollution using simple, relatable examples.
4. Involve the students through short questions on conservation.
5. Solve an exercise question on global warming together with the class.
6. Summarize the lesson and highlight the key points of Man and His Environment.

Classroom Activities

• Group discussion: students compare food chains with pollution.

• Board activity: solve a question on conservation in pairs.

• Quick quiz on the key terms of Man and His Environment.

Assessment Questions

1. Define ecosystem.
2. Differentiate between food chains and pollution.
3. Explain with an example: conservation.
4. Write a short note on global warming.

Homework

• Read Man and His Environment from the book and solve the exercise questions.

• Prepare short notes on ecosystem and food chains.

Time Allocation

4–6 periods of 40 minutes each